

ShopMate: A Low-Cost Vision-Based Autonomous Following Robot Using ESP32-CAM and Arduino Uno for Retail and Warehouse Assistance

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Abstract—Repetitive material transportation and the physical burden of carrying products in retail and warehouse environments present persistent operational challenges that motivate the development of affordable autonomous robotic solutions. This paper presents the design and implementation of a low-cost vision-based autonomous following robot utilizing an ESP32-CAM module and Arduino Uno microcontroller, eliminating dependency on computationally intensive platforms such as Raspberry Pi, ROS, OpenCV, LIDAR, or deep learning frameworks. A central contribution of this work is a custom dual-color fiducial marker—a blue triangular region embedding a red circular dot—coupled with a lightweight two-stage embedded validation algorithm performing real-time RGB565 pixel analysis directly on the ESP32-CAM. This dual-color spatial verification strategy substantially reduces false positive detections prevalent in conventional single-color tracking systems, enabling reliable target discrimination within visually complex indoor environments. UART-based inter-device communication facilitates responsive directional command transmission, while PWM-controlled four-wheel drive ensures precise robotic navigation. Experimental evaluation confirmed an effective detection range of 20–70 cm, frame processing latency of 15–25 ms, zero false positive rate, and motor response time under 30 ms—demonstrating practical real-time autonomous following performance using purely microcontroller-level hardware. The proposed system offers a financially accessible and deployable robotic assistance solution for retail customer support and warehouse inventory operations, establishing a compelling foundation for future low-cost embedded robotics research.

Index Terms—Autonomous following robot, ESP32-CAM, fiducial marker, embedded computer vision, RGB565, Arduino Uno, retail robotics, warehouse automation, low-cost robotics.

I. INTRODUCTION

The increasing adoption of automation technologies in retail, warehouse, and service industries has accelerated the development of autonomous mobile robotic systems designed to improve operational efficiency, reduce manual labor, and enhance customer convenience. In modern retail stores and warehouse facilities, employees are routinely engaged in repetitive material transportation tasks—such as moving products from storage areas to display shelves or delivery zones. Simultaneously, customers in large shopping environments often experience physical inconvenience while carrying products throughout extended shopping sessions. These challenges have

created growing interest in intelligent robotic assistance systems capable of autonomously supporting both employees and customers in indoor commercial environments.

Recent advances in embedded systems, computer vision, and low-power microcontrollers have enabled the development of compact autonomous robots for object tracking, navigation, and human-following applications. Autonomous following robots are increasingly being explored for industrial logistics, hospital assistance, smart transportation, inventory management, and customer service automation. However, the majority of existing human-following robotic systems depend heavily on computationally expensive hardware platforms such as Raspberry Pi, NVIDIA Jetson modules, LIDAR sensors, depth cameras, and Robot Operating System (ROS)-based frameworks combined with OpenCV or deep learning algorithms for image processing and target recognition. Although these approaches deliver high tracking accuracy and robust environmental perception, they significantly increase system cost, power consumption, and implementation complexity. Consequently, such solutions are often impractical for small and medium-scale retail businesses where affordability, simplicity, and low maintenance are critical deployment requirements.

In addition to hardware complexity, conventional color-based target tracking systems frequently suffer from poor reliability in visually dynamic environments. Many low-cost systems rely on single-color blob detection techniques, causing the robot to incorrectly follow unrelated objects containing similar colors—such as clothing, packaging materials, advertisements, or store decorations. Such false detections substantially reduce the practical usability of autonomous following systems in real-world retail environments where multiple colored objects coexist within the camera’s field of view. Achieving reliable target discrimination using computationally lightweight embedded vision techniques therefore remains an important research challenge for low-cost autonomous robotic platforms.

To address these limitations, this project proposes a low-cost vision-based autonomous following robot using an ESP32-

CAM module and Arduino Uno microcontroller, without relying on OpenCV, ROS, machine learning models, or external processing hardware. The proposed system introduces a custom dual-color fiducial marker composed of a blue triangular region with a centrally positioned red circular dot, designed to improve target uniqueness and reduce false positive detections. A lightweight two-stage marker validation algorithm is implemented directly on the ESP32-CAM using raw RGB565 pixel analysis. In the first stage, the system identifies a sufficiently large blue pixel region and computes its centroid position. In the second stage, the existence of red pixels is verified within a localized search window surrounding the computed centroid. Only when both validation conditions are satisfied does the system classify the object as a valid tracking target. This dual-color spatial verification strategy enables reliable marker recognition while maintaining extremely low computational overhead suitable for resource-constrained embedded hardware.

The primary objective of this research is to design and implement an economically accessible autonomous robotic assistant capable of real-time human-following behavior in indoor retail and warehouse environments. The system aims to demonstrate that embedded computer vision using low-cost microcontroller platforms can achieve practical autonomous tracking functionality without requiring high-performance computing devices. Additional objectives include implementing reliable UART-based inter-device communication between the ESP32-CAM and Arduino Uno, developing PWM-based directional motor control for four-wheel robotic navigation, and experimentally validating system performance in practical retail-oriented application scenarios.

The proposed robot targets two major application domains. In warehouse and retail inventory management environments, the robot functions as an autonomous product transportation assistant that follows employees carrying the designated marker while transporting products to target shelves. In customer-oriented retail scenarios, the robot operates as a hands-free autonomous shopping assistant capable of following customers while carrying purchased items. These scenarios demonstrate the potential of affordable autonomous robotics to improve workplace productivity, reduce physical workload, and enhance customer experience in commercial environments.

The scope of this work is primarily focused on indoor marker-based autonomous following under controlled lighting conditions. The system performs directional navigation based on real-time marker localization within the camera frame and does not incorporate advanced simultaneous localization and mapping (SLAM), obstacle avoidance, or machine learning-based environmental perception. Nevertheless, the proposed approach demonstrates that reliable autonomous following behavior can be achieved through efficient low-level image processing and embedded system optimization.

Several prior studies have contributed significantly to this field. Chien et al. developed a multiple interface-based firefighting robot integrating embedded control and re-

mote interaction mechanisms for hazardous environments [1]. Bruce et al. introduced a fast and computationally efficient color image segmentation technique for interactive robotic systems using real-time color blob tracking [2]. Cui et al. investigated color-based visual servoing methods for target following and mobile robot navigation [3]. Sani and Karimian further demonstrated autonomous navigation and tracking using marker-based vision systems combined with inertial sensing techniques [4]. While these studies established important foundations for autonomous robotic vision systems, many existing approaches continue to rely on comparatively expensive processing platforms or computationally demanding algorithms. In contrast, the proposed system emphasizes affordability, simplicity, low computational complexity, and practical deployment feasibility while maintaining reliable real-time tracking performance for retail and warehouse automation applications.

II. SYSTEM DESIGN AND METHODOLOGY

A. System Architecture Overview

The proposed autonomous following robot is founded upon a carefully partitioned dual-processor embedded architecture, in which two microcontroller units operate in close coordination to deliver seamless real-time vision-based tracking and motor-driven navigation. Rather than concentrating all processing responsibilities onto a single high-cost platform, the system deliberately distributes computational tasks between two specialized units—the ESP32-CAM and the Arduino Uno—each optimized for its assigned role. This architectural separation ensures that the computationally intensive demands of live image acquisition and pixel-level analysis do not contend with the timing-sensitive requirements of motor actuation, thereby improving both system responsiveness and operational stability under real-world conditions.

As illustrated in Fig. 1, the ESP32-CAM module serves as the primary vision and decision-making unit of the system. It is responsible for continuously capturing live camera frames, executing the dual-stage marker detection algorithm, computing positional directional decisions, and transmitting command signals to the secondary controller. The Arduino Uno, operating as the dedicated low-level motor control unit, receives these directional instructions and drives the four DC gear motors through an L298N dual H-bridge motor driver to produce the appropriate physical movement of the robotic chassis.

The complete operational pipeline flows unidirectionally from visual perception to physical actuation. The ESP32-CAM acquires a raw image frame from its onboard OV2640 camera sensor and processes it through a two-stage pixel-level validation algorithm. Upon confirming a valid tracking target, the system encodes a single-character directional command and dispatches it to the Arduino Uno over a UART serial link. The Arduino then decodes the received command and activates the corresponding motor drive configuration through the L298N driver. As depicted in Fig. 3, this end-to-end

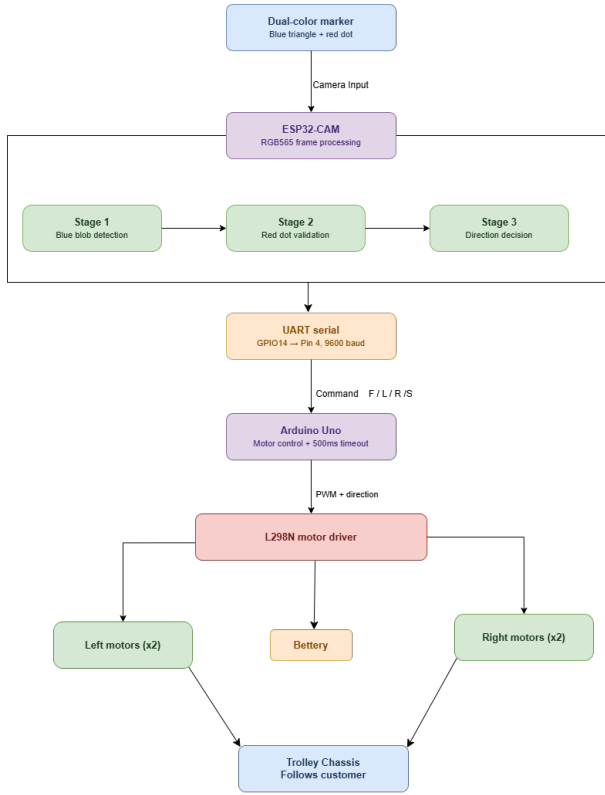


Fig. 1. System Block Diagram showing the hardware architecture and signal flow from visual input to physical locomotion.

data flow—from raw camera input through image processing, serial communication, motor control logic, and wheel actuation—forms a continuous closed-loop cycle that repeats autonomously throughout system operation, enabling the robot to track and follow the designated marker holder in real time without any human intervention or external computing infrastructure.

B. Custom Dual-Color Fiducial Marker Design

A fundamental and distinguishing innovation of this system is the design of a custom dual-color fiducial marker that serves as the exclusive visual identification target for the autonomous following robot. Unlike conventional human-following systems that attempt to track skin tone, clothing color, or infrared signals, the proposed approach relies on a purpose-engineered physical marker that introduces a deliberate two-color spatial constraint, making reliable embedded detection feasible even on severely resource-constrained hardware.

The marker consists of a blue equilateral triangle, approximately 8–10 cm in base width and 7–9 cm in height, constructed by printing or drawing on rigid cardboard. At the precise geometric center of this triangular region, a red circular dot of approximately 1.5–2 cm in diameter is positioned. The physical dimensions of the marker were carefully selected to ensure adequate visibility within the ESP32-CAM's effective detection range while keeping the pixel footprint manageable for the QQVGA (160 × 120) resolution image processing

pipeline. During testing, the marker was either held directly by a user or mounted onto a shopping basket to simulate realistic retail and warehouse operation scenarios.

The dual-color spatial configuration of the marker is the cornerstone of the system's false-detection immunity. The detection logic enforces two sequential conditions that must both be satisfied before the robot is permitted to move:

- **Blue-only object:** A plain blue object—such as a garment, packaging, or store decoration—will satisfy the first detection stage by accumulating sufficient blue pixels but will invariably fail the second stage due to the complete absence of a centrally located red region.
- **Red-only object:** A standalone red object present anywhere in the frame will never trigger the initial blue blob detection and will therefore be entirely ignored by the system.
- **Valid marker:** Only the precise physical combination of a large blue region with a red dot geometrically centered within it will pass both validation stages and authorize robot movement.

This design strategy renders the system highly robust against false detections in visually complex retail environments where multiple colored objects, advertisements, packaging materials, and clothing items coexist simultaneously within the camera's field of view.

C. Two-Stage Marker Detection Algorithm

The marker detection and directional decision logic is implemented entirely within the ESP32-CAM firmware and operates through three sequential processing stages during each camera frame cycle. The complete algorithmic flow is depicted in Fig. 2.

1) *Stage 1—Blue Blob Detection:* At the beginning of each processing cycle, the ESP32-CAM captures a single image frame at QQVGA resolution (160 × 120 pixels) encoded in RGB565 format. In this format, each pixel is represented as a 16-bit unsigned integer where the upper 5 bits encode the red channel, the middle 6 bits encode the green channel, and the lower 5 bits encode the blue channel. To recover meaningful 8-bit color channel values suitable for threshold-based comparison, each pixel undergoes bitwise extraction and left-shift scaling as follows:

$$R = ((\text{pixel} \gg 11) \& 0x1F) \ll 3 \quad (1)$$

$$G = ((\text{pixel} \gg 5) \& 0x3F) \ll 2 \quad (2)$$

$$B = (\text{pixel} \& 0x1F) \ll 3 \quad (3)$$

Following channel extraction, each pixel is evaluated against a set of blue dominance conditions:

$$B > R + 30 \quad (4)$$

$$B > G + 20 \quad (5)$$

$$B > 80 \quad (6)$$

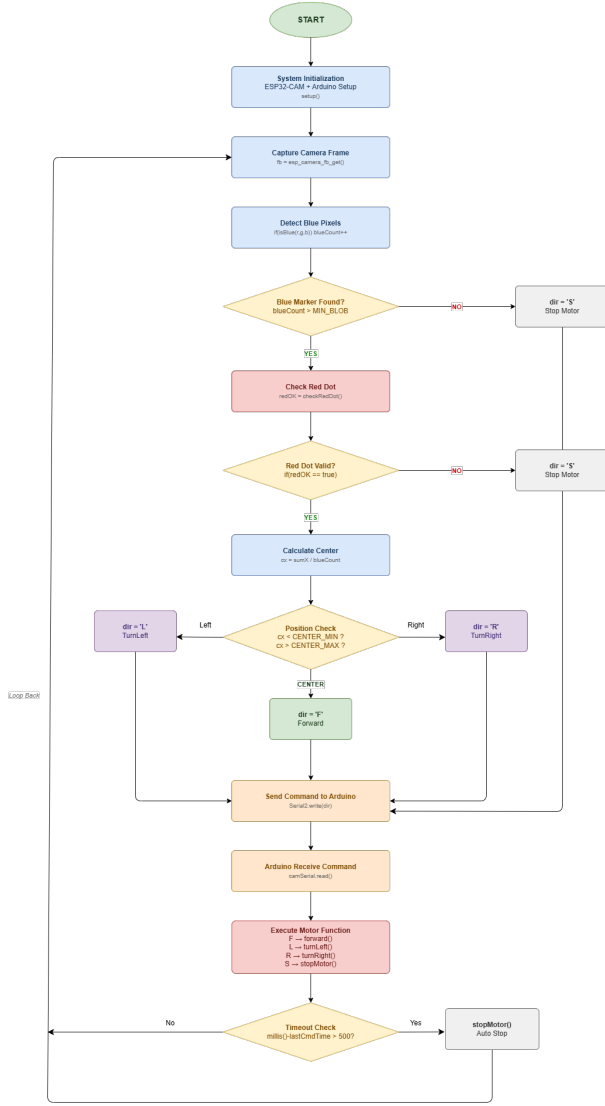


Fig. 2. System Flowchart illustrating the complete runtime execution logic from system startup through closed-loop marker tracking, directional decision-making, motor actuation, and safety timeout enforcement.

All pixels satisfying conditions (4)–(6) simultaneously are classified as blue, and their x and y coordinates are accumulated. The total blue pixel count (blueCount) and cumulative coordinate sums are used to compute the blob centroid:

$$C_x = \frac{\sum x_{\text{blue}}}{\text{blueCount}}, \quad C_y = \frac{\sum y_{\text{blue}}}{\text{blueCount}} \quad (7)$$

To prevent erroneous detections caused by image noise, minor color reflections, or excessively large uniform blue backgrounds, a blob size validation filter is applied. Detections with fewer than 200 blue pixels are discarded as noise, while detections exceeding 8,000 blue pixels are rejected as oversized or ambiguous regions. Only blobs within this valid range proceed to Stage 2.

2) *Stage 2—Red Dot Centroid Confirmation:* Upon establishing a valid blue blob centroid at coordinates (C_x, C_y) ,

TABLE I
DIRECTIONAL COMMAND MAPPING BASED ON CENTROID POSITION

C_x Position	Command	Motor Action
$C_x < 50$	L	Left turn
$50 \leq C_x \leq 110$	F	Forward
$C_x > 110$	R	Right turn
No valid marker	S	Stop

the algorithm proceeds to verify the spatial presence of red pixels at the center of the detected blue region. A constrained 40×40 pixel search window is defined symmetrically around the computed centroid, bounded by $(C_x \pm 20, C_y \pm 20)$, and clipped to the frame boundaries to prevent memory access violations.

Within this localized window, each pixel is independently evaluated against a set of red dominance thresholds:

$$R > 160 \quad (8)$$

$$R > G + 60 \quad (9)$$

$$R > B + 60 \quad (10)$$

The number of pixels satisfying all three red conditions is accumulated as redCount. If $\text{redCount} > 30$, the presence of the red dot at the centroid of the blue region is confirmed and the marker is classified as a valid tracking target. If the red pixel count falls below this threshold, the detection is rejected and a Stop command is issued regardless of the blue blob result. This centroid-anchored spatial verification ensures that the red region must physically coincide with the geometric center of the blue blob, precisely matching the spatial layout of the custom fiducial marker and eliminating false confirmations from scattered red objects located elsewhere in the scene.

3) *Stage 3—Directional Command Generation:* Once a valid marker detection is confirmed through both preceding stages, the system determines the appropriate directional movement command based solely on the horizontal position of the blue blob centroid (C_x) within the 160-pixel wide camera frame. The frame is partitioned into three horizontal zones with the command mapping defined in Table I.

This positional zone partitioning effectively implements a proportional steering correction mechanism. When the marker drifts toward the left edge of the frame, the robot turns left to realign; when it drifts to the right, the robot turns right; when the marker remains horizontally centered, the robot advances forward. This simple yet effective threshold-based approach provides stable real-time directional alignment between the robot and the tracked marker without requiring complex control algorithms such as PID controllers or Kalman filters, maintaining minimal computational overhead on the resource-constrained embedded platform.

D. Communication Architecture and Protocol

Reliable inter-device communication between the ESP32-CAM and the Arduino Uno is achieved through a UART-based

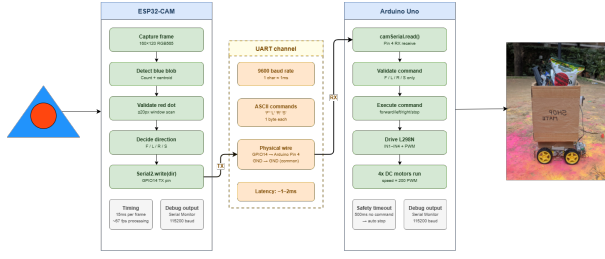


Fig. 3. Communication and Data Flow Diagram illustrating the end-to-end pipeline from image acquisition through UART transmission to motor actuation.

asynchronous serial link, as illustrated in Fig. 3. Following each frame processing cycle, the ESP32-CAM encodes the computed directional decision as a single ASCII character and transmits it through GPIO14 (configured as the UART TX pin) to Arduino Uno Pin 4, which is configured as a SoftwareSerial receive pin. Communication operates at a baud rate of 9600bps, and since each directional command occupies exactly one byte, the resulting transmission latency is approximately 1–2 ms per command—a duration negligible relative to the 15–25 ms frame processing cycle and therefore introducing no perceptible delay in robot response behavior.

The four valid command characters and their corresponding motor behaviors are:

```
'F' -> Forward (both motor pairs drive forward)
'L' -> Left turn (right forward, left reverse/stop)
'R' -> Right turn (left forward, right reverse/stop)
'S' -> Stop (all motors halted)
```

A parallel debug output channel operates simultaneously on the ESP32-CAM at 115200 baud through the primary hardware serial port, streaming real-time diagnostic information during development and testing. A representative sample of this debug output is shown below:

```
Blue:1200 CX:85 CY:62 Red:YES DIR:F
Blue:980 CX:35 CY:58 Red:YES DIR:L
Blue:1050 CX:128 CY:60 Red:YES DIR:R
Blue:0 CX:-1 CY:-1 Red:NO DIR:S
Blue:650 CX:80 CY:55 Red:NO DIR:S
```

A critical safety mechanism is embedded within the Arduino Uno’s control logic. If no valid directional command character is received within a predefined 500 ms timeout window, the Arduino automatically halts all motors, as shown below:

```
CMD: F <- forward motion
CMD: L <- left turn
CMD: R <- right turn
TIMEOUT - stop <- no command within 500ms
```

This failsafe behavior prevents uncontrolled or runaway robot motion in the event of ESP32-CAM processing failure, UART signal interruption, sudden marker loss, or power instability, significantly enhancing operational safety during real-world deployment.

E. Hardware Components

The hardware architecture of the proposed system was assembled entirely from commercially available low-cost embed-

TABLE II
HARDWARE COMPONENT SPECIFICATIONS

Component	Specification	Qty
ESP32-CAM	AI-Thinker, OV2640 sensor	1
Arduino Uno	ATmega328P, 16 MHz	1
L298N Motor Driver	Dual H-Bridge, 2 A/channel	1
DC Gear Motor	3–6 V, with wheel	4
Li-ion Battery	3.7 V nominal	2
4WD Robotic Chassis	Standard four-wheel drive platform	1
Custom Fiducial Marker	Blue triangle with red center dot	1

TABLE III
ESP32-CAM TO ARDUINO UNO CONNECTIONS

ESP32-CAM Pin	Arduino Pin	Function
GPIO14 (UART TX)	Pin 4 (SoftwareSerial RX)	Directional command transmission
GND	GND	Common ground reference

ded components, as summarized in Table II. Each component was selected based on its cost-effectiveness, availability, power compatibility, and suitability for real-time embedded vision and motor control applications.

The ESP32-CAM module integrates an Espressif ESP32-S system-on-chip with an onboard OV2640 image sensor, 4 MB of flash memory, and built-in Wi-Fi and Bluetooth capabilities. For this project, only the camera and UART communication functionalities are utilized; wireless features remain inactive to conserve power and reduce firmware complexity. The Arduino Uno, based on the ATmega328P microcontroller operating at 16 MHz with 2 KB of SRAM and 32 KB of flash memory, provides sufficient resources for serial command reception, motor control logic, and timeout management. The L298N dual H-bridge motor driver module supports bidirectional control of up to four DC motors with independent PWM speed regulation on two motor channels, making it well-suited for differential steering of the four-wheel robotic platform.

F. Wiring and Pin Configuration

All hardware modules are electrically interconnected through carefully defined pin assignments to ensure reliable UART communication, motor actuation, and a stable common electrical reference across the entire system. The complete wiring configurations are presented in Tables III and IV.

The two 3.7 V Li-ion batteries are connected in series to produce a nominal 7.4 V supply voltage for the L298N motor driver and the four DC gear motors. The ESP32-CAM module is powered independently through its dedicated USB interface during testing and development phases to guarantee stable camera operation and uninterrupted serial debugging without voltage fluctuations caused by motor load variations. All ground terminals across the ESP32-CAM, Arduino Uno, L298N motor driver, and battery supply are interconnected at a single common node, which is essential for maintaining UART signal integrity and preventing ground loop interference between the vision processing and motor control subsystems.

TABLE IV
ARDUINO UNO TO L298N MOTOR DRIVER CONNECTIONS

Arduino Pin	L298N Pin	Function
Pin 8	IN1	Left motor direction control
Pin 9	IN2	Left motor direction control
Pin 10	IN3	Right motor direction control
Pin 11	IN4	Right motor direction control
Pin 5 (PWM)	ENA	Left motor speed control
Pin 6 (PWM)	ENB	Right motor speed control
GND	GND (common)	Common ground

G. Working Principle and End-to-End Data Flow

The complete system operation follows a deterministic, continuously repeating closed-loop cycle that integrates visual perception, algorithmic decision-making, serial communication, and physical motor actuation into a seamless autonomous following behavior.

Upon powering on, both the ESP32-CAM and Arduino Uno execute their respective initialization routines—configuring the camera sensor, establishing UART communication channels, and setting motor driver pins to their default inactive states. The ESP32-CAM then enters a continuous frame acquisition loop, capturing 160×120 pixel RGB565 images from the OV2640 sensor at approximately 40–67 frames per second depending on scene complexity and processing load.

Each captured frame is immediately subjected to the two-stage marker detection pipeline. The full pixel array is scanned to identify and accumulate blue pixels meeting the dominance and brightness thresholds, and the blue blob centroid is computed from the qualifying pixels. If the blob size falls within the valid range of 200–8,000 pixels, the algorithm proceeds to scan the 40×40 pixel centroid window for red pixel confirmation. Based on the combined outcome of both stages, a single directional command character is selected and transmitted to the Arduino Uno via UART within the same processing cycle.

Upon receiving a valid command character, the Arduino Uno immediately maps it to the corresponding motor actuation function—driving both motor channels forward for F, reversing the left channel for R, reversing the right channel for L, or halting all channels for S—and applies a configured PWM speed value of 200 to the enable pins to regulate motor velocity. The timeout watchdog runs concurrently, resetting on each valid command reception and triggering an automatic motor stop if 500 ms elapses without a new command. This complete cycle—from frame capture through detection, communication, and motor response—executes repeatedly and autonomously, enabling the robot to continuously track and follow the designated dual-color marker holder in real time under normal indoor operating conditions.

III. IMPLEMENTATION AND RESULTS

A. Prototype Implementation and System Setup

The proposed autonomous following robot was implemented using a low-cost embedded hardware architecture

consisting of an ESP32-CAM module, Arduino Uno micro-controller, L298N dual H-bridge motor driver, and a four-wheel drive robotic chassis platform. The complete prototype was designed to evaluate the feasibility of real-time embedded vision-based human-following behavior without relying on high-performance computing devices or external image processing frameworks.

The ESP32-CAM module served as the primary vision processing and decision-making unit of the system. It was mounted at the front-center position of the robotic chassis at an approximate height of 15 cm from the ground surface to maximize the forward field of view during marker tracking. The onboard OV2640 camera sensor was oriented horizontally in the forward direction to continuously capture image frames from the surrounding environment. The camera was configured to operate in QQVGA resolution (160×120 pixels) using the RGB565 image format to minimize memory consumption and improve real-time processing speed on the resource-constrained embedded processor.

The Arduino Uno functioned as the low-level motor control unit responsible for receiving directional commands from the ESP32-CAM and controlling the movement of the four DC gear motors. An L298N dual H-bridge motor driver module was used to provide sufficient current amplification and bidirectional control for the motors. PWM-based speed control signals were generated through the Arduino Uno to regulate motor operation during forward movement and directional turning.

The robotic platform was powered using two 3.7 V lithium-ion batteries connected in series to provide stable voltage for the motor driver and DC motors. During the experimental phase, the ESP32-CAM module was powered separately through its USB interface to ensure stable camera operation and uninterrupted serial debugging. All ground terminals of the ESP32-CAM, Arduino Uno, motor driver, and battery supply were interconnected to establish a common electrical reference and ensure reliable UART communication between the embedded devices.

A custom dual-color fiducial marker was developed as the visual target for autonomous following. The marker consisted of a blue triangular region with a centrally positioned red circular dot. The marker dimensions were selected to remain sufficiently visible within the effective operating range of the camera while maintaining low computational processing overhead. The marker was either held by a user or attached to a shopping basket during testing procedures to simulate practical retail and warehouse operation scenarios.

B. Experimental Testing Procedure

To evaluate the effectiveness and reliability of the proposed system, experimental testing was conducted in multiple progressive stages under controlled indoor lighting conditions. The testing methodology focused on validating the vision detection algorithm, UART communication reliability, directional motor control, and overall autonomous following performance.

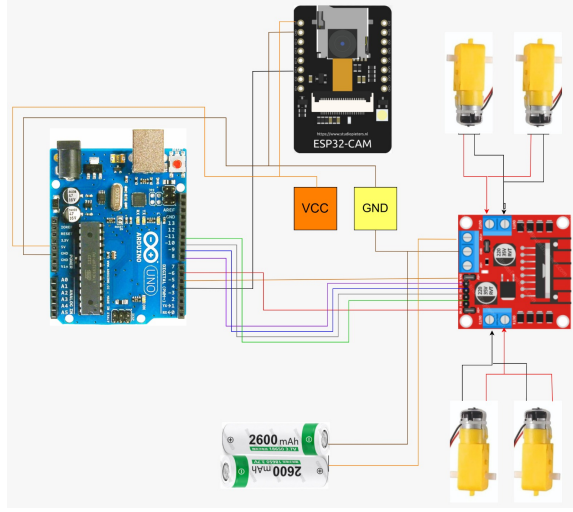


Fig. 4. Complete prototype hardware setup including the ESP32-CAM vision module, Arduino Uno controller, L298N motor driver, power supply arrangement, and robotic chassis assembly.

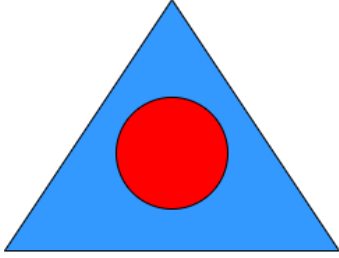


Fig. 5. Custom dual-color fiducial marker: blue equilateral triangle with centrally positioned red circular dot.

1) *Stage 1—Vision System Validation:* The first testing stage focused on validating the embedded image processing algorithm implemented on the ESP32-CAM module. The ESP32-CAM was connected to a personal computer through USB communication, and real-time debugging information was monitored using the Serial Monitor operating at 115200 baud. The custom fiducial marker was positioned at different horizontal locations and distances relative to the camera frame to evaluate blue blob detection, centroid computation accuracy, and red-dot validation reliability. Representative serial outputs observed during experimental testing are shown below:

```
Blue:1200 CX:85 CY:62 Red:YES DIR:F
Blue:980 CX:35 CY:58 Red:YES DIR:L
Blue:1050 CX:128 CY:60 Red:YES DIR:R
Blue:0 CX:-1 CY:-1 Red:NO DIR:S
Blue:650 CX:80 CY:55 Red:NO DIR:S
```

The experimental observations confirmed that the system successfully differentiated the valid dual-color marker from unrelated blue-colored objects in the environment. Objects containing only blue coloration without the red centroid confirmation consistently failed the second validation stage and did not trigger robot movement, demonstrating the effectiveness

TABLE V
EXPERIMENTAL DETECTION AND FOLLOWING RESULTS

Test Condition	Blue Count	Red Valid.	Direction	Response
Marker centered, 30 cm	~1200	YES	F	Correct
Marker left	~980	YES	L	Correct
Marker right	~1050	YES	R	Correct
No marker	0	NO	S	Correct
Other blue object	~650	NO	S	No false follow
Marker at 20 cm	~2800	YES	F	Correct
Marker at 70 cm	~300	YES	F	Correct
Marker partial occlusion	~150	NO	S	Correct stop

of the proposed dual-color validation strategy.

2) *Stage 2—UART Communication and Motor Control Validation:* The second testing stage evaluated the reliability of UART-based serial communication between the ESP32-CAM and Arduino Uno. Single-character directional commands were transmitted from the ESP32-CAM through GPIO14 at 9600 baud, and the Arduino Uno displayed the received outputs on the Serial Monitor for validation:

```
CMD: F
CMD: L
CMD: R
TIMEOUT - stop
```

A timeout-based safety mechanism was additionally validated. If no valid command was received within 500 ms, the system automatically stopped all motors—confirming reliable failsafe behavior under communication failure or marker loss conditions.

3) *Stage 3—Full System Integration and Autonomous Following Evaluation:* The final experimental stage involved complete system integration testing under practical operating conditions. During testing, a user carried the custom dual-color marker while walking in multiple directions within the camera's field of view. The robot continuously analyzed incoming image frames, determined the relative marker position, and generated appropriate directional movement commands in real time. When the marker was intentionally removed, partially occluded, or replaced with unrelated blue objects, the system immediately halted robot movement, confirming the effectiveness of the dual-stage validation mechanism in preventing false tracking behavior.

C. Experimental Results and Performance Evaluation

The proposed system demonstrated reliable real-time operation throughout all experimental stages. Table V summarizes the primary experimental results obtained during marker detection and autonomous following evaluation.

The results indicate that the proposed dual-color validation strategy effectively eliminated false positive detections caused by unrelated blue-colored objects. The robot successfully followed the designated marker within an effective operating distance of approximately 20–70 cm. Marker detection performance degraded slightly under severe partial occlusion conditions due to insufficient blue pixel accumulation; however,

TABLE VI
SYSTEM PERFORMANCE METRICS

Performance Metric	Measured Value
Frame processing time	15–25 ms
UART transmission latency	1–2 ms
Effective detection range	20–70 cm
False positive rate	0% (controlled env.)
False negative rate	<5%
Motor response time	<30 ms

this behavior also prevented unintended tracking of incomplete targets. System performance metrics are summarized in Table VI.

The observed processing speed demonstrates that the ESP32-CAM is capable of performing lightweight embedded computer vision tasks in real time despite its limited computational resources. The average frame processing time remained sufficiently low to maintain responsive robot movement, while UART communication latency between the ESP32-CAM and Arduino Uno was negligible relative to the overall system response time. Overall experimental results confirm that the proposed system successfully achieved the primary objectives of low-cost autonomous following, reliable target validation, and real-time embedded vision-based navigation using only microcontroller-level hardware platforms.

IV. CONCLUSION

This research successfully demonstrated the design and implementation of a low-cost vision-based autonomous following robot using an ESP32-CAM module and Arduino Uno for retail and warehouse assistance applications. The developed system achieved reliable real-time target tracking and autonomous following behavior without relying on computationally expensive platforms such as Raspberry Pi, OpenCV, ROS, LIDAR, or deep learning frameworks. By utilizing lightweight embedded image processing directly on the ESP32-CAM, the project established that practical autonomous robotic assistance can be achieved using resource-constrained microcontroller hardware with minimal system cost and complexity.

A major contribution of this work is the development of a custom dual-color fiducial marker and a corresponding two-stage validation algorithm for robust target identification. Unlike conventional single-color tracking approaches that are vulnerable to false detections in visually complex environments, the proposed system combines blue blob detection with centroid-based red dot verification to ensure accurate target discrimination. Experimental evaluation confirmed that the robot could reliably distinguish the intended marker from unrelated colored objects and generate appropriate directional responses in real time. The achieved detection performance, low communication latency, and responsive motor control validate the effectiveness of the proposed embedded vision strategy for autonomous following applications.

The prototype successfully fulfilled all primary project objectives, including real-time marker detection using RGB565

pixel analysis, UART-based communication between the ESP32-CAM and Arduino Uno, PWM-controlled four-wheel robotic navigation, and practical validation under realistic indoor operating conditions. The system demonstrated stable operation within an effective range of approximately 20–70 cm while maintaining negligible false positive detections during controlled testing. Furthermore, the modular hardware architecture and software design allow straightforward customization of detection thresholds, movement logic, and operational parameters for deployment in different environments.

From a practical perspective, the proposed robot presents significant potential for real-world commercial applications. In warehouse and retail environments, the system can function as an autonomous product transportation assistant that reduces the physical workload of employees during inventory restocking and product delivery operations. In customer-oriented scenarios, the robot may serve as a hands-free autonomous shopping assistant capable of following customers while carrying purchased items. Such applications have the potential to improve workplace efficiency, reduce repetitive manual labor, and enhance user convenience, particularly for elderly individuals or customers carrying heavy products. The affordability of the proposed system further increases its suitability for small and medium-scale businesses that cannot adopt high-cost robotic automation solutions.

Although the proposed system achieved successful autonomous following behavior, several limitations remain that can be addressed in future work. The current implementation primarily operates under controlled indoor lighting conditions and does not incorporate advanced obstacle avoidance or environmental mapping capabilities. Future improvements may include the integration of PID-based motor control for smoother directional movement, adaptive thresholding techniques for improved detection under varying illumination conditions, distance estimation using blob-size analysis for proportional speed control, and ultrasonic or infrared sensors for obstacle avoidance. Additionally, future research may explore lightweight machine learning or neural network-based detection techniques to further improve robustness and scalability while maintaining low hardware requirements.

Overall, this work demonstrates that reliable autonomous following behavior can be achieved through efficient embedded computer vision techniques using inexpensive microcontroller-based hardware platforms. The proposed approach contributes toward the development of affordable autonomous robotic assistance systems suitable for practical deployment in retail, warehouse, and service-oriented environments, while also providing a foundation for future research in low-cost embedded robotics and intelligent automation technologies.

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